

第三章 矩阵的进一步讨论 (2)

知识点 可逆矩阵、伴随矩阵、逆矩阵的计算、初等变换求逆矩阵.

课后练习

1. 判别题:

- 1) 初等矩阵都可逆. (✓)
- 2) 初等矩阵的逆矩阵也是初等矩阵. (✓)
- 3) 已知 $AB = AC$, 其中 A 是可逆阵, 则 $B = C$ (✓)
- 4) 已知 $AB = O$, 其中 $A \neq O$, 则 $B = O$ (✗)
若 A 可逆, 则 $B = O$
- 5) 设 n 阶方阵 A 与 B 等价, 则 $|A| = 0$ 当且仅当 $|B| = 0$ (✓)

2. 填空题:

1) 已知 $\lambda_i \neq 0 (i=1,2,3)$, 则 $\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}^{-1} = \begin{pmatrix} \lambda_1^{-1} & 0 & 0 \\ 0 & \lambda_2^{-1} & 0 \\ 0 & 0 & \lambda_3^{-1} \end{pmatrix}$; $\begin{pmatrix} 0 & 0 & \lambda_1 \\ 0 & \lambda_2 & 0 \\ \lambda_3 & 0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & 0 & \lambda_1^{-1} \\ 0 & \lambda_2^{-1} & 0 \\ \lambda_3^{-1} & 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$; $\begin{pmatrix} 5 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 5 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -2 & 0 & 0 \\ -2 & 5 & 0 & 0 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & -5 & 8 \end{pmatrix}$

$\begin{pmatrix} 8 & 3 \\ 5 & 2 \end{pmatrix}^{-1} = \frac{1}{-1} \begin{pmatrix} 2 & -3 \\ -5 & 8 \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ 5 & -8 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 & \frac{1}{3} \\ 2 & 1 & 0 \\ 4 & 3 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -2 & 1 \\ 3 & 0 & 0 \end{pmatrix}$

2) 已知矩阵 $A_{3 \times 4}$ 的秩为 2, $B = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, 则 $\text{rank}(A^T B) = \text{rank}(A^T) = 2$

3) 设矩阵 $A = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, 则 $A^{-1} = \begin{pmatrix} -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$; $(A^T)^{-1} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 0 & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$

伴随矩阵 $(3A)^* = 3A^* = \begin{pmatrix} 18 & -36 & 0 \\ -18 & 0 & 0 \\ 0 & 0 & -18 \end{pmatrix}$; $(2A^*)^{-1} = \frac{1}{2}(A^*)^{-1} = \frac{1}{2}A^{-1} = -\frac{1}{8} \begin{pmatrix} 0 & 2 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{4} & 0 \\ -\frac{1}{8} & -\frac{1}{8} & 0 \\ 0 & 0 & -\frac{1}{4} \end{pmatrix}$

4) 已知 $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, 则行列式 $|A|$ 中所有元素的代数余子式之和为 7

5) 设 A, B 均为 n 阶可逆矩阵, 则下列各式中不正确的是 (B C D) (多选)

A. $(2A)^T = 2A^T$ B. $(3A)^{-1} = 3A^{-1}$ C. $(A^{-1})^T = (A^T)^{-1}$ D. $(A+B)^T = A^T + B^T$

E. $(A+B)^{-1} = A^{-1} + B^{-1}$ F. $(AB)^{-1} = B^{-1}A^{-1}$ G. $(AB)^T = B^T A^T$

$(A+B)(A^{-1}+B^{-1}) = \dots$

$$\textcircled{1} AP_1 = B$$

$$\textcircled{2} P_2 B = E$$

6) 设 A 是 3 阶矩阵, 将 A 的第 2 列加到第 1 列得矩阵 B , 再将矩阵 B 的第 2 行

与第 3 行交换得单位矩阵, 记 $P_1 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $P_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, 则 (D)

A. $A = P_1 P_2$

B. $A = P_1^{-1} P_2$

C. $A = P_2 P_1$

D. $A = P_2 P_1^{-1} \Rightarrow A = P_2^{-1} P_1^{-1}$

3. 已知 A 是 3 阶矩阵, $|A| = \frac{1}{2}$, 求 $|(2A)^{-1} - 5A^*|$.

$$\begin{aligned} |(2A)^{-1} - 5A^*| &= |E - 10AA^*| = |-4E| = (-4)^3 = -64 \\ &= |2A| | (2A)^{-1} - 5A^* | = 2^3 |A| | (2A)^{-1} - 5A^* | \\ &\Rightarrow |(2A)^{-1} - 5A^*| = -16 \end{aligned}$$

4. 已知 $A = (a_{ij})$ 是 3 阶非零矩阵, A_{ij} 为元素 a_{ij} 的代数余子式, 若 $a_{ij} + A_{ij} = 0$,

求 A 的行列式 $|A|$.

$$\begin{aligned} \text{由题知 } A_{ij} &= -a_{ij} \Rightarrow \text{伴随矩阵 } A^* = (A_{ji}) = (-a_{ji}) = -A^T \\ \text{又 } AA^* &= |A|E \Rightarrow -AA^T = |A|E \end{aligned}$$

$$\text{两边取行列式} \Rightarrow |-A| \cdot |A| = |A|^3 \Rightarrow (-1)^3 |A|^2 = |A|^3 \Rightarrow |A| = 0 \text{ 或 } -1$$

5. 已知 A 为三阶方阵, ξ_1, ξ_2, ξ_3 为三维列向量, $A\xi_1 = \xi_1 + \xi_2 + \xi_3, A\xi_2 = \xi_1 + \xi_2,$

$A\xi_3 = \xi_1$, 若矩阵 $P = (\xi_1, \xi_2, \xi_3)$ 可逆, 求 $P^{-1}AP$.

$$\Rightarrow A(\xi_1, \xi_2, \xi_3) = (\xi_1, \xi_2, \xi_3) \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \Rightarrow P^{-1}AP = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

6. 求解矩阵方程 $AX = 2X + B$, 其中 $A = \begin{pmatrix} 0 & 1 & 1 \\ -2 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, B = \begin{pmatrix} 0 & 3 \\ -4 & -2 \\ 0 & -2 \end{pmatrix}$.

$$\Rightarrow (A - 2E)X = B \quad |A - 2E| = \begin{vmatrix} -2 & 1 & 1 \\ -2 & -2 & 0 \\ 0 & 0 & 0 \end{vmatrix} = 0 \neq 0$$

$$X = (A - 2E)^{-1}B = \frac{1}{|A - 2E|} (A - 2E)^* B$$

$$\begin{aligned} (A - 2E)^* &= \begin{pmatrix} -2 & 1 & 1 \\ -2 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ (A - 2E)X &= B \Rightarrow \left(\begin{array}{ccc|cc} -2 & 1 & 1 & 0 & 3 \\ -2 & -2 & 0 & -4 & -2 \\ 0 & 0 & 0 & 0 & -2 \end{array} \right) \xrightarrow[r_3 \div (-2)]{r_2 - r_1} \left(\begin{array}{ccc|cc} -2 & 1 & 1 & 0 & 3 \\ 0 & -3 & -1 & -4 & -5 \\ 0 & 0 & 0 & 0 & -2 \end{array} \right) \\ &\xrightarrow[r_3 \div (-2)]{r_2 + 3r_3} \left(\begin{array}{ccc|cc} -2 & 1 & 1 & 0 & 3 \\ 0 & -3 & -1 & -4 & -5 \\ 0 & 0 & 0 & 0 & -2 \end{array} \right) \xrightarrow[r_3 \div (-2)]{r_2 \div (-3)} \left(\begin{array}{ccc|cc} -2 & 1 & 1 & 0 & 3 \\ 0 & 1 & \frac{1}{3} & \frac{4}{3} & \frac{5}{3} \\ 0 & 0 & 0 & 0 & -2 \end{array} \right) \\ &\xrightarrow[r_2 \div (-1)]{r_1 - r_2} \left(\begin{array}{ccc|cc} -2 & 0 & \frac{2}{3} & -\frac{4}{3} & \frac{1}{3} \\ 0 & 1 & \frac{1}{3} & \frac{4}{3} & \frac{5}{3} \\ 0 & 0 & 0 & 0 & -2 \end{array} \right) \xrightarrow[r_2 \div (-1)]{r_1 \div (-2)} \left(\begin{array}{ccc|cc} 1 & 0 & \frac{1}{3} & \frac{2}{3} & -\frac{1}{6} \\ 0 & 1 & \frac{1}{3} & \frac{4}{3} & \frac{5}{3} \\ 0 & 0 & 0 & 0 & -2 \end{array} \right) \Rightarrow X = \begin{pmatrix} \frac{1}{3} & 0 \\ \frac{2}{3} & \frac{1}{3} \\ 0 & -\frac{1}{3} \end{pmatrix} \end{aligned}$$

7. 设矩阵 $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, 矩阵 B 满足 $ABA^* = 2BA^* + E$, 其中 A^* 为伴随矩阵, E 是单位矩阵,

求矩阵 B .

$$\begin{aligned} |A| &= 3 \neq 0 \Rightarrow A, A^* \text{ 可逆且 } (A^*)^{-1} = \frac{A}{|A|} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ |A - 2E| &= \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{vmatrix} = 1 \neq 0 \Rightarrow A - 2E \text{ 可逆.} \\ \text{且 } (A - 2E)^{-1} &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} B &= (A - 2E)^{-1} E (A^*)^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 \\ 0 & 0 & -\frac{1}{3} \end{pmatrix} \\ &= E(1, 2) \cdot E(3, -1) \end{aligned}$$

$$(A-E)BA^{-1}=3E$$

8. 已知矩阵 A 的伴随矩阵 $A^* = \text{diag}(1, 1, 1, 8)$, 且 $ABA^{-1} = BA^{-1} + 3E$, 求 B .

解 $|A^*| = 8 \neq 0 \Rightarrow A^*$ 可逆

假设 $|A| = 0$, 则有 $AA^* = |A|E = 0 \Rightarrow A = 0(A^*)^{-1} = 0$, 此时 $A^* = 0$ 矛盾

故 $|A| \neq 0 \Rightarrow |A||A^*| = |A|E \Rightarrow |A^*| = |A|^3 \therefore |A| = 2$

由题设知 $(A-E)BA^{-1} = 3E$

$$A^*(A-E)B = 3A^*A$$

$$(|A|E - A^*)B = 6E$$

$$B = 6 \cdot (2E - A^*)^{-1} = \text{diag}(6, 6, 6, -1)$$

9. 已知 $AP = P\Lambda$, 其中 $P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}$, $\Lambda = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix}$, 求 $\varphi(A) = A^8(5E - 6A + A^2)$.

解 $|P| = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & -3 \\ 1 & -1 & 0 \end{vmatrix} = 3 \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -6 \neq 0 \therefore P$ 可逆且 $P^{-1} = -\frac{1}{6} \begin{pmatrix} -2 & -2 & -2 \\ -3 & 0 & 3 \\ -1 & 2 & -1 \end{pmatrix}$

$$A = P\Lambda P^{-1} \Rightarrow A^k = P\Lambda^k P^{-1}$$

$$\varphi(A) = 5A^8 - 6A^9 + A^{10} = P \cdot 5\Lambda^8 P^{-1} - P \cdot 6\Lambda^9 P^{-1} + P \cdot \Lambda^{10} P^{-1}$$

$$= P(5\Lambda^8 - 6\Lambda^9 + \Lambda^{10})P^{-1}$$

$$= P(5E - 6\Lambda + \Lambda^2)\Lambda^8 P^{-1} = P \begin{pmatrix} 12 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} P^{-1} = \begin{pmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{pmatrix}$$

10. 已知 $A = P^{-1}BP$, $f(x)$ 是多项式, 证明: $f(A) = P^{-1}f(B)P$.

证明: 设 $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

$$A = P^{-1}BP \quad f(A) = a_n A^n + a_{n-1} A^{n-1} + \dots + a_1 A + a_0 E$$

$$A^k = \underbrace{P^{-1}BP P^{-1}BP \dots P^{-1}BP}_{k \text{ 次}} = a_n P^{-1}B^n P + a_{n-1} P^{-1}B^{n-1} P + \dots + a_1 P^{-1}BP + a_0 P^{-1}EP$$

$$= P^{-1}B^k P \quad = P^{-1}(a_n B^n + a_{n-1} B^{n-1} + \dots + a_1 B + a_0 E)P = P^{-1}f(B)P \quad \square$$

11. 设 n 阶矩阵 A 的伴随矩阵为 A^* , 证明: (1) 若 $|A| = 0$, 则 $|A^*| = 0$; (2) $|A^*| = |A|^{n-1}$.

(1) 反证: 假设 $|A^*| \neq 0 \Rightarrow A^*$ 可逆, (2) 当 $|A| = 0$ 时 $\Rightarrow |A^*| = 0 = |A|^{n-1}$

又由 $AA^* = |A|E = 0$ 知

$A = 0$, 从而 $A^* = 0$ 矛盾

故当 $|A| = 0$ 时, $|A^*| = 0$

当 $|A| \neq 0$ 时, 对 $AA^* = |A|E$ 两边取行列式

$$\Rightarrow |A| \cdot |A^*| = |A|E = |A|^n \cdot 1$$

$\therefore |A| \neq 0 \Rightarrow |A^*| = |A|^{n-1} \quad \square$

12. 已知 n 阶矩阵 A, B 满足 $A+B=AB$. 证明: (1) $A-E$ 可逆; (2) $AB=BA$.

(1) $AB-A-B=0 \Rightarrow (A-E)(B-E)=E$

$\therefore A-E$ 可逆且 $(A-E)^{-1} = B-E$

(2) 由于 $(A-E)^{-1} = B-E$

从而 $E = (B-E)(A-E) = BA - A - B + E \Rightarrow BA = A + B = AB \quad \square$